Alexandria Engineering Journal 115 (2025) 47–56

Contents lists available at ScienceDirect

Alexandria Engineering Journal

journal homepage: www.elsevier.com/locate/aej

Original article

Optimizing UAV-assisted IoT sensor networks: A multi-objective approach to data collection and routing

Yasir I. Mohammed [a], Rosilah Hassan [a], Mohammad Kamrul Hasan [a,*], Huda Saleh Abbas [b], Muhammad Attique Khan [c,f], Jamel Baili [d], Deepak Gupta [e,g]

[a] Center for Cyber Security, Faculty of information Science and Technology, Universiti Kebangsaan Malaysia (UKM), Bangi 43600, Malaysia
[b] Department of Computer Science, College of Computer Science and Engineering, Taibah University, Saudi Arabia
[c] Department of Computer Science and Engineering, College of Informatics, Korea University, 145 Anam-ro, Seongbuk-gu, Seoul 02841, Republic of Korea
[d] Department of Computer Engineering, College of Computer Science, King Khalid University, Abha 61413, Saudi Arabia
[e] Maharaja Agrasen Institute of Technology, Delhi, India
[f] Department of AI, College of Computer Engineering and Science, Prince Mohammad Bin Fahd University, KSA
[g] Centre for Research Impact & Outcome, Chitkara University Institute of Engineering and Technology, Chitkara University, Punjab, India

ARTICLE INFO

Keywords:
Data gathering
Q-learning
UAVs
FANET
Reinforcement learning

ABSTRACT

Unmanned Aerial Vehicles (UAVs) are powerful tools for data gathering in IoT sensor networks, enhancing efficiency in challenging environments. This research proposes an optimized UAV deployment and data routing approach, addressing energy use, coverage, and communication delays. The study guarantees comprehensive coverage and effective data management by employing sophisticated optimisation methodologies. The solution achieves significant energy savings, reduced latency, and improved performance, offering a robust framework for large-scale IoT applications like agriculture, disaster response, and environmental monitoring.

1. Introduction

Data gathering using unmanned aerial vehicles (UAVs), known as drones, has revolutionized how information is collected in diverse environments. With advanced sensors such as high-resolution cameras, LiDAR systems, thermal sensors, and multi-spectral imagers, UAVs provide unparalleled access to remote, hazardous, or otherwise inaccessible areas [1,2]. These technologies enable the collection of diverse data types, including aerial imagery, detailed 3D terrain models, and environmental metrics such as temperature, humidity, and gas concentrations [3,4]. Applications span numerous industries: in agriculture, UAVs monitor crop health and optimize irrigation; in construction and infrastructure management, they assess structural integrity; in environmental monitoring, they track wildlife, deforestation, and natural disasters; and in disaster response, they provide real-time intelligence for rescue operations[5,6]. By combining autonomy, speed, and precision, UAVs offer significant cost and time savings while enhancing data collection accuracy compared to traditional methods [7,8].

Integrating UAVs with emerging technologies such as the Internet of Things (IoT) has expanded their potential, introducing advanced data collection and routing capabilities. Recent advancements focus on leveraging UAVs as mobile data collectors within IoT sensor networks, addressing challenges such as dynamic network topology, energy constraints, and latency in communication[9,10]. Despite these benefits, deploying UAVs in IoT environments requires solving complex optimization problems related to efficient coverage, energy consumption, and communication reliability [11]. Traditional methods struggle to balance these competing factors, particularly in real-time, large-scale deployments [12].

Building upon this foundational framework, the subsequent sections delve into advanced methodologies for optimizing UAV-assisted IoT networks. Multi-objective optimization techniques, including NSGA-II and NSGA-III, are utilized to determine optimal UAV deployment strategies that maximize coverage while minimizing energy consumption and operational costs. Simultaneously, reinforcement learning algorithms, such as Q-Learning, dynamically optimize data routing paths, enabling real-time adaptability to evolving network conditions [13]. These methodologies address practical and theoretical challenges, providing a robust framework for scalable and sustainable UAV-IoT integration [14].

* Corresponding author.
E-mail addresses: hasankamrul@ieee.org (M.K. Hasan), habbas@taibahu.edu.sa (H.S. Abbas), attique.khan@ieee.org (M.A. Khan), Jabaili@kku.edu.sa (J. Baili), drdeepakgupta.cse@gmail.com (D. Gupta).

https://doi.org/10.1016/j.aej.2024.12.018
Received 27 October 2024; Received in revised form 24 November 2024; Accepted 5 December 2024
Available online 11 December 2024
1110-0168/© 2024 The Author(s). Published by Elsevier B.V. on behalf of Faculty of Engineering, Alexandria University. This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/).

As IoT sensor networks become increasingly pervasive across industries like intelligent agriculture, environmental monitoring, disaster management, and urban planning, scalability, energy efficiency, and latency challenges become even more critical[15]. The paragraphs detail how advanced optimization techniques and adaptive algorithms can revolutionize UAV-assisted IoT networks, developing scalable and energy-efficient solutions to meet the growing demands of modern sensor deployments [16]. This research contributes to a comprehensive strategy for integrating UAVs into IoT systems, enabling enhanced data collection capabilities, improved network performance, and more resilient infrastructure for real-time monitoring applications. The research paper contributions can be presented as below:

- Introduces a systematic optimization approach for UAV-assisted IoT networks, focusing on two interconnected sub-problems: data gathering and routing;
- Utilizes multi-objective optimization algorithms (NSGA-II, NSGA-III) to determine optimal UAV deployment patterns, addressing coverage, energy efficiency, cost, and other key factors;
- Integrates reinforcement learning with a Q-Learning algorithm for adaptive routing, optimizing data transmission paths in a dynamic network;
- Proposes a problem decomposition strategy separating UAV deployment and routing while preserving their interdependence, enhancing overall optimization;
- Provides a comprehensive evaluation framework, comparing performance against baseline strategies, incorporating visualizations and statistical analysis;

The article's structure is as follows: Section 2 reviews relevant prior studies, providing a comprehensive overview of the existing literature. Section 3 delves into the proposed methodology, detailing its framework and implementation. Section 4 presents the experimental results, accompanied by an in-depth analysis and critical discussion. Lastly, Section 5 offers the conclusion, summarizing the essential findings and highlighting the contributions of this research.

2. Related work

The research study explores a novel multi-objective optimization framework for managing the multi-depot vehicle routing problem with limited supply (MDVRPLS) in the context of refined oil distribution [16]. The study integrates three core objectives minimizing operational costs and overtime penalties while maximizing station satisfaction to address the challenges posed by oil shortages. By leveraging a robust optimization approach, the authors account for demand uncertainties and propose a multi-objective particle swarm optimization (MOPSO) algorithm to efficiently solve the MDVRPLS problem. Numerical experiments validate the robustness of the model, demonstrating its superiority over traditional methods by effectively balancing conflicting objectives and mitigating uncertainties. The study offers significant implications for resource allocation and decision-making in scenarios of scarce supply, contributing to the broader field of optimization in logistics and supply chain management.

The study presents a groundbreaking approach to address the limitations of conventional Medium Access Control (MAC) protocols in Flying Ad Hoc Networks (FANETs) with Unmanned Aerial Vehicles (UAVs) [17]. By integrating intelligent sensing and communication mechanisms with Multiple-Input Multiple-Output (MIMO) technology, the authors propose a CSMA/CA-based MAC protocol that leverages Space-Division Multiple Access (SDMA). This innovation significantly enhances the network's ability to handle dynamic topologies and high mobility, overcoming the constraints of traditional access schemes like TDMA and FDMA. The inclusion of a maximum likelihood estimation model for real-time channel assessment ensures efficient resource utilization and minimizes access delays. Additionally, the study introduces a fairness-oriented adaptive scheduling algorithm utilizing the Age of Information (AoI) metric, optimizing data freshness and prioritizing critical transmissions. Simulation results validate the protocol's superior throughput and reduced latency, making it an adaptable solution for future large-scale FANETs in complex scenarios.

The research study introduces a novel approach to enhancing the trajectory tracking and prediction of micro-unmanned aerial vehicles (micro-UAVs) using advanced radar technology[18]. The study framework integrates spatial-temporal data through a 4D MIMO radar system, addressing the challenges posed by the low Radar Cross-Section (RCS) and the presence of noise in detection environments. Unlike traditional methods that separate detection and tracking processes, this framework jointly optimizes both tasks to achieve superior results in low Signal-To-Noise Ratio (SNR) conditions. Furthermore, it incorporates a transformer-based prediction model that leverages both local and global spatial-temporal dependencies for trajectory prediction. Experimental validations, including simulations and real-world scenarios, demonstrate that the STIF outperforms existing methodologies in terms of accuracy and robustness, even under adverse conditions. The research represents a significant advancement in UAV trajectory tracking technology, emphasizing its applicability in safety-critical applications such as urban airspace monitoring.

The study presents a sophisticated approach for vehicle detection and tracking in aerial images[19]. The study leverages centroid tracking and template matching techniques to enhance detection robustness and precision across varied environmental conditions and viewing angles. The authors incorporate advanced image pre-processing methods, such as contrast-limited adaptive histogram equalization (CLAHE), to improve image quality and feature extraction. Additionally, the YOLOv4 backbone, augmented with spatial pyramid pooling (SPP) and cross-stage partial networks (CSP), significantly optimizes real-time detection performance. The system not only achieves high accuracy in traffic monitoring but also demonstrates potential applications in urban planning and disaster management by efficiently geo-referencing detected vehicles. Despite its advancements, challenges such as computational intensity and sensitivity to occlusions remain. Overall, the work represents a substantial contribution to the field of computer vision, offering practical implications for real-world aerial surveillance and traffic systems.

This research study investigates the integration of UAVs with Wireless Sensor Networks (WSNs) for efficient data collection in IoT platforms[20,21]. The study addresses critical challenges such as optimizing UAV flight paths to minimize energy consumption, enhancing data collection efficiency, and improving overall system performance. The study proposes a novel hybrid optimization model based on the Traveling Salesman Problem (TSP) and Bezier curve smoothing, termed BL-TSP. This approach ensures that UAVs follow the shortest, smoothest paths while accounting for motion constraints such as turning radius and altitude. A cost function is formulated to evaluate UAV trajectories, incorporating variables like path length, fuel consumption, and power usage [22]. The model demonstrates improved data collection rates and energy efficiency by employing the Kalman filter for data aggregation and the Lyapunov function for UAV control. The research contributes significantly to the field by optimizing UAV-assisted data collection in IoT frameworks, offering potential applications in environmental monitoring, disaster management, and other large-scale sensor network deployments.

This research study presents an innovative approach to optimizing the placement and operation of a fleet of flying ferries for data gathering in WSNs [23]. The study addresses the critical issue of energy efficiency and balanced resource utilization in a ferry-assisted WSN system, which is particularly crucial for extending the system's operational lifetime. The research introduces a novel energy-balanced ferry fleet placement scheme, FEB, which ensures fair task allocation, energy efficiency, and optimal ferry deployment. This optimization problem is modelled as a mixed-integer linear programming (MILP) problem, solved through a

two-stage process. The first stage employs a linear sum assignment problem (LSAP) to manage the one-to-one matching between sensors and ferries.

In contrast, the second stage utilizes a genetic algorithm to solve the Travelling Salesman Problem (TSP), optimizing the ferries' trajectories. The study also leverages the powered-Voronoi diagram for dynamic region partitioning, adjusting service areas based on the residual energy of each ferry. Simulation results demonstrate that the proposed FEB scheme outperforms existing methods, such as K-Means and Spiral, regarding energy consumption, fairness in energy utilization, and overall system lifetime, significantly contributing to WSN data gathering optimization.

The research presents an optimization framework for three-dimensional (3D) trajectory design and data collection scheduling for fixed-wing UAVs, aiming to enhance energy efficiency and security[24]. The authors introduce a novel propulsion energy model for 3D UAV flights, addressing gaps in previous models focused on two-dimensional trajectories. Using a Difference of Convex (DC) and Successive Convex Approximation (SCA) based algorithm, the study optimizes the UAV's trajectory and sensor node transmission schedules to minimize energy consumption while ensuring data security against eavesdroppers. Simulations demonstrate that the 3D trajectory model achieves significant energy savings compared to conventional methods.

This study introduces an energy-efficient framework for UAV-assisted data collection in WSNs [25]. The research focuses on minimizing energy consumption by optimizing UAV stop positions, sensor assignments, and travel paths. The problem is formulated as a non-convex optimization challenge, tackled through decomposition into three sub-problems: UAV positioning, sensor clustering, and path planning using the Traveling Salesman Problem (TSP). The proposed iterative method significantly reduces energy consumption compared to existing approaches, enhancing the efficiency of UAV-based data collection in delay-tolerant WSN applications.

This research study addresses the problem of optimizing the fine-grained trajectories of multiple UAVs for efficient data gathering from WSNs [26]. The increasing adoption of UAVs as mobile data collectors offers significant advantages due to their high manoeuvrability, flexibility, and speed. However, complexity increases when multiple UAVs are deployed collaboratively, requiring the optimization of their flight paths and detailed hovering and travelling strategies in continuous three-dimensional space. The study introduces the Fine-Grained Trajectory Plan (FTP) problem, which aims to minimize the maximum flight time of UAVs while ensuring the complete collection of sensory data from the WSN. The authors propose a mathematical model for this NP-hard problem alongside approximation algorithms guaranteeing constant-factor performance. By exploring single and multi-UAV scenarios, the research contributes a robust framework for UAV trajectory planning, optimizing data collection in complex and challenging environments. Extensive simulations are employed to validate the efficacy of the proposed algorithms, offering insights into their practical application in real-world scenarios involving WSNs[27].

3. Methodology

3.1. NSGA-II

The NSGA-II is a widely used optimization algorithm for solving multi-objective optimization problems. It extends classical genetic algorithms by incorporating mechanisms to handle multiple objectives simultaneously, aiming to find a set of trade-off solutions rather than a single optimal solution. NSGA-II achieves this by balancing convergence towards the Pareto-optimal front and diversity across the solutions. Here is a detailed explanation and analysis of NSGA-II:

1) Core Principles and Non-Dominated Sorting

NSGA-II uses the concept of dominance to classify solutions into different fronts. A solution is said to dominate another if it is at least as good in all objectives and strictly better in at least one. The algorithm starts by generating an initial population and then classifies individuals into non-dominated fronts through a non-dominated sorting process. Any other solution does not dominate solutions on the first front, while those on the second front are dominated only by solutions on the first front, and so on. Each front is assigned a rank, with the first being the most desirable. This ranking forms the foundation for selection during evolutionary operations.

NSGA-II begins by organizing a population of N solutions based on dominance relationships. A solution $x_i$dominates another solution $x_i(x_i \prec \quad x_j)$ If:

$$\forall k \quad \in \quad \{1,2,\ldots.,M\}, \quad \mathscr{F}_k(x_i) \leq \mathscr{F}_k(x_j) \tag{1}$$

$$\exists k \in \{1,2,\ldots,M\}, \mathscr{F}_k(x_i) < \mathscr{F}_k(x_j) \tag{2}$$

where $\mathscr{F}_k(x)$ is the k-th objective function. Solutions are grouped into non-dominated fronts F_1,F_2 such that F_1 contains all non-dominated solutions, F_2 contains solutions dominated only by those in F_1, and so on. Each front is assigned a rank $R(x)$where $R(x) = 1 for x \in F_1, R(x) = 2 for x \in F_2$ and so forth

2) Crowding Distance and Diversity Preservation

One of the critical challenges in multi-objective optimization is maintaining diversity in the population. NSGA-II introduces a crowding distance metric to quantify solutions' proximity to their neighbours within a Pareto front. The algorithm favours solutions in less crowded areas, ensuring the population remains diverse across the Pareto front. This crowding distance prevents premature convergence and allows the algorithm to explore the objective space effectively. The combination of ranking and crowding distance forms the basis of NSGA-II's elitist selection mechanism, where solutions are selected based on their front rank and crowding distance.

3) Selection, Crossover, and Mutation

NSGA-II employs standard genetic operators such as binary tournament selection, crossover, and mutation to generate offspring. During selection, pairs of solutions compete based on their front rank, and if they belong to the same front, the one with a higher crowding distance is chosen. Ensures better solutions (lower-ranked fronts) and diverse solutions (higher crowding distance) are favoured. The offspring generated through crossover and mutation undergo evaluation, and the combined parent-offspring population is then sorted for the next generation. The algorithm uses a steady-state strategy where the population size is fixed, and the best solutions are retained across generations. NSGA-II uses binary tournament selection based on rank and crowding distance. A solution x_i is preferred over x_j if

$$R(x_i) \quad < R(x_j) or (R(x_i) = R(x_j) and d(x_i) > d(x_j)) \tag{3}$$

Selected parents undergo crossover and mutation to generate offspring. A commonly used crossover is the simulated binary crossover (SBX), governed by:

$$x_i^{new} = \frac{1}{2}\left[(1+\beta)x_i + (1-\beta)x_j\right], \quad x_i^{new} = \frac{1}{2}\left[(1-\beta)x_i + (1+\beta)x_j\right] \tag{4}$$

Where β depends on a distribution index η_c mutation is often performed using polynomial mutation:

$$x_i^{mutated} = x_i + \Delta \tag{5}$$

Where Δ is a perturbation based on the mutation distribution index η_m

4) Advantages, Limitations, and Applications

NSGA-II has several advantages, including its computational efficiency (owing to the $O(MN^2)$ complexity of non-dominated sorting, where M is the number of objectives, and N is the population size), its ability to maintain diversity, and its simplicity in implementation. It has been successfully applied in various engineering design, scheduling, and machine learning domains. However, it also has limitations. For instance, as the number of objectives increases (in many objective problems), the performance of NSGA-II may degrade due to difficulties preserving diversity and handling high-dimensional objective spaces. Advanced variants, such as NSGA-III, have been developed to address these issues. Despite these challenges, NSGA-II remains a cornerstone algorithm in multi-objective optimization and is widely used for benchmarking new techniques.

3.2. Q-learning

Q-learning is a model-free reinforcement learning algorithm that enables an agent to learn an optimal action-selection policy for a given environment. It is rooted in **Q-values**, or **state-action values**, which represent the expected utility of taking a specific action a in a particular state s, and following the optimal policy. The main goal of Q-learning is to iteratively update these Q-values until they converge to the optimal values, denoted as $Q^* \quad (s,a)$. This algorithm does not require a model of the environment and can operate in stochastic settings.

The Q-learning update equation is central to the algorithm. Given a state s, an action a, the resulting reward r, the next state s', and a discount factor $\gamma \in [0,1]$, the Q-value is updated as follows:

$$Q(s,a) \leftarrow Q(s,a) + \alpha[r + \gamma \max \quad Q(s',a') - Q(s,a)] \tag{6}$$

Where

$Q(s,a)$ is the current estimate of the Q-value for state s and action a.

a is the learning rate $(0 < a \leq 1)$, determining how much newly acquired information overrides the old information.

$r + \gamma \max_{a'} Q(s',a')$ is the target, combining the immediate reward r and the discounted maximum future reward obtainable from state s'

The term $r + \gamma \max_{a'} Q(s',a') - Q(s,a)$ is the temporal difference (TD) **error**, which measures the difference between the predicted Q-value and the target.

Q-learning guarantees convergence to the optimal Q-values $Q^*(s,a)$ under certain conditions, such as using a suitable exploration strategy, sufficient exploration of the state-action space, and a decaying learning rate α alphaα. The algorithm balances exploration (trying new actions to gather information) and exploitation (choosing actions that maximize the current estimate of Q-values). The Bellman optimality equation serves as the theoretical foundation, ensuring that the Q-values eventually satisfy:

$$Q^*(s,a) = \mathbb{E}[r + \gamma \max_{a'} \quad Q^*(s',a')] \tag{7}$$

This recursive definition enables Q-learning to backpropagate the rewards from future states to previous states without requiring prior knowledge of the environment's transition probabilities.

While Q-learning is powerful, it faces several practical challenges. For large state or action spaces, maintaining a Q-table becomes infeasible due to memory and computation constraints. Function approximators like deep neural networks are used in such cases, leading to algorithms like Deep Q-Networks (DQN). Additionally, hyperparameters such as a, γ , and the exploration strategy significantly affect the algorithm's performance and must be carefully tuned. The algorithm can struggle in environments with delayed rewards or sparse reward landscapes. Despite these challenges, Q-learning remains a foundational algorithm in reinforcement learning, illustrating how agents can learn optimal policies through trial-and-error interaction with their environment.

3.3. General Problem Formulation

In a specified environment E, the problem entails designing an optimal data forwarding mechanism within a network comprising UAVs, sensors, and base stations. Each UAV u in the set U follows a predetermined trajectory. W_u, covering a zone C_u where it can gather data from sensors S and communicate with other UAVs and base stations B, the challenge lies in enabling each UAV to intelligently select its next hop, either another UAV or a base station within its coverage for forwarding collected data. This mechanism must maximize the quantity of data packets successfully received by the base stations while minimizing the delay in the data forwarding process. The solution seeks to leverage cooperative strategies among UAVs to enhance the efficiency and reliability of data transmission in dynamic, UAV-assisted sensor networks, potentially applicable in diverse areas like environmental monitoring and disaster management. Mathematically, we define the following variables:

1) **Sensor Nodes**
   Position of sensor $s_i : (x_i, y_i)$, the Data generation rate of the sensor $s_i : \lambda_i$, the Data size of the sensor $s_i : d_i$
2) **UAVs**
   Position of UAV u_j at time $t : \left(x_j(t), y_j(t)\right)$, the Energy level of UAV $u_j : E_j(t)$, Communication range of UAV $u_j : R_j$, Speed of UAV $u_j : v_j$, Flight path of UAV $u_j : W_j(t)$
3) **Routes:** Set of possible routes for each UAV $u_j : \mathscr{R}_j$, Next hop selection variable: h_{ij} indicating if UAV u_j moves to sensor s_i next.

The objective is to maximize the quantity of data packets D_{received} successfully received by the base stations while minimizing the delay Δ_{total} in the data forwarding process. It can be expressed as a multi-objective optimization as:

$$\begin{aligned} &\text{Optimize} && [\max D_{\text{received}}, \min \Delta_{\text{total}}] \\ &\text{Subject to} && E_j(t) \geq E_{\min}, \\ & && \sqrt{\left(x_j(t) - x_i\right)^2 + \left(y_j(t) - y_i\right)^2} \leq R_j, \\ & && \sum_{i \in S} d_i h_{ij} \leq C_j \\ & && h_{ij} \in \{0,1\}.. \end{aligned} \tag{8}$$

Where:

- $D_{\text{received}} = \sum_{j=1}^{m} \sum_{i=1}^{n} \lambda_i h_{ij}$
- $\Delta_{\text{total}} = \sum_{j=1}^{m} \int_0^T \delta_j(t) dt \delta_j(t)$ represents the delay function for UAV u_j.

The optimization process aims to find the optimal next hop h_{ij} For each UAV, efficient data collection and forwarding are ensured by leveraging cooperative strategies among UAVs. This approach enhances the efficiency and reliability of data transmission in dynamic, UAV-assisted sensor networks, with potential applications in environmental monitoring and disaster management.

The provided graph in Fig. 1. Represents a WSN environment with UAVs, base stations, sensors, and rendezvous points. The graph is plotted on a grid, which helps visualize various elements' spatial distribution and communication range. Here is a detailed explanation of the components depicted in the graph:

a) The UAVs are represented by blue triangles (Δ). Each UAV follows a predefined trajectory and operates within a particular coverage zone, depicted by circles around them. These circles indicate the communication range of the UAVs.
b) Purple squares represent the base stations. Base stations act as fixed points within the network where data collected by UAVs is ultimately transmitted. They have a significant role in data aggregation and communication with external networks.

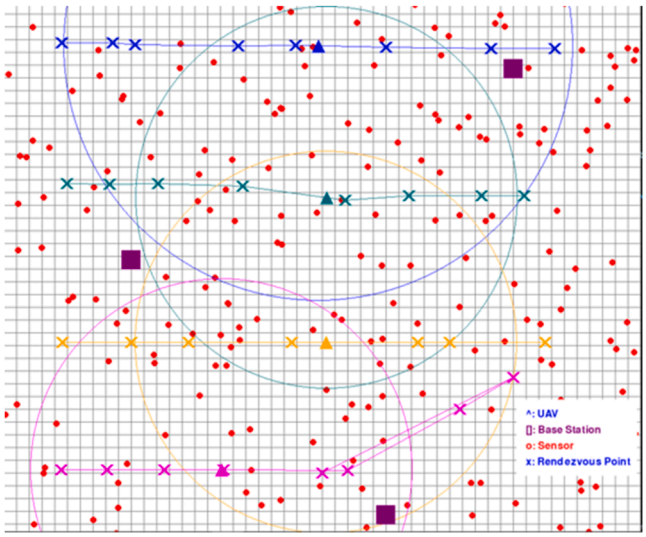

Fig. 1. Graphical representation of a WSN environment with UAVs, base stations, sensors, and rendezvous points, illustrating communication ranges and trajectories.

c) Sensors are represented by red circles (0). Sensors are distributed across the environment to collect various types of data. These sensors generate data that needs to be collected and forwarded by the UAVs to the base stations.
d) Rendezvous points are represented by coloured crosses (×). UAVs can meet at these points to exchange data or make routing decisions. They are strategically placed to optimize data forwarding and reduce delays.
e) Communication range is depicted by circles around each UAV, showing the area within which a UAV can communicate with sensors, other UAVs, and potentially base stations. These circles are crucial for understanding the coverage area of each UAV and planning the data collection routes accordingly.
f) Trajectories are indicated by lines connecting the UAVs and the rendezvous points. Different colours might represent different UAVs or different paths taken by the UAVs.

The graph illustrates a complex network where UAVs collect sensor data and forward it to base stations. The UAVs must efficiently manage their routes and make decisions on the next hop, whether it be another UAV or directly to a base station. The rendezvous points provide strategic locations for data exchange and coordination among UAVs. The optimization challenge here involves:

3.4. Data generation model

The data are generated in each sensor using two probabilistic models; the first one is the typical distribution model, which provides the number of packets at a particular time step, and the second one is the exponential distribution model. It gives the time interval from one moment of generation to the next moment of generation. To formulate this process mathematically, we can consider two variables: the number of packets generated at a given time step and the interval between successive packet generations. Let's denote these variables as follows:

$N(t)$ Represents the number of packets generated at time step t. This variable follows a normal distribution, which can be mathematically expressed as $N(t) \sim N(\mu, \sigma^2)$. Here, μ and σ^2 represent the mean and variance of the distribution, respectively. The normal distribution is appropriate for modelling phenomena where most observations cluster around a central value (mean), and probabilities for values further away from the mean taper off symmetrically in both directions.

T_i Denotes the time interval between the generation of packets at two successive moments, say i and $i+1$. This variable follows an exponential distribution, which is mathematically expressed as $T_i \sim Exp(\lambda)$. In this expression, λ is the rate parameter of the exponential distribution. The exponential distribution is commonly used to model the time until an event occurs and is particularly useful in scenarios where events happen independently and at a constant average rate.

Thus, the complete mathematical model for your sensor data generation process can be formulated as follows:

$$N(t) \sim N(\mu, \sigma^2), Ti \sim Exp(\lambda) \tag{9}$$

In this model, $N(t)$ and T_i are independent random variables. The standard distribution component models the packet generation at each time step, while the exponential distribution component models the time interval between these generations. It's important to note that the parameters μ, σ^2, and λ need to be determined based on empirical data or domain-specific knowledge to reflect the behaviour of the sensor network accurately

3.5. UAV mobility model

To mathematically formalize the mobility models for UAVs in the specified environment E, we need to define parameters that encapsulate the movement dynamics of each UAV in the network. Each UAV, denoted as u in the set U, follows a pre-determined trajectory W_u. This trajectory is a critical component in the design of the data forwarding mechanism. Let us define the following parameters:

1) Start Point $P_{u,start}$. The starting coordinates of the UAV u's trajectory are presented in the equation. Depending on the environment's complexity, it can be represented as a vector in a 2D or 3D space. We consider 2D space because the environment is assumed to be described as a horizontal plane containing the sensors. For simplicity in (3), let's consider a 2D plane.

$$P_{u,start} = \{x_{u,start}, y_{u,start}\} \tag{10}$$

2) End Point, $P_{u,end}$ The ending coordinates of the UAV u's trajectory on the same plane. Referring to (4).

$$P_{u,end} = \{x_{u,end}, y_{u,end}\} \tag{11}$$

3) Trajectory, W_u The trajectory can be represented as a sequence of waypoints or a parametric equation that the UAV follows from $P_{u,start}$ to $P_{u,end}$. We consider that the trajectory is a straight line. Referring as:

$$W_u(t) = P_{u,start} + t \cdot (P_{u,end} - P_{u,start}), t \in [0, 1] \tag{12}$$

4) Coverage Zone, Cvg_u It is the area where UAVs u can gather data from sensors and communicate with other UAVs and base stations. It can be defined as a circle (or sphere in 3D) with a radius. r_u centred at the UAV's current position $W_u(t)$
5) Cycle, C_u: Assuming the UAV repeats its trajectory, the cycle can be defined as the time it takes for UAV u to complete one entire trajectory from
$P_{u,start} to P_{u,end}$ and back to $P_{u,start}$.
6) The mathematical model for each UAV's movement can thus be described as follows:

Each UAV u in the set U starts from $P_{u,start}$ follows the trajectory W_u, and ends at $P_{u,end}$ with the cycle C_u: determining the periodicity of this movement. The coverage zone Cvg_u at any point in time is a function of the UAV's current position on its trajectory.

The UAV's movement, coupled with its coverage zone, defines its

ability to interact with sensors S, other UAVs, and base stations B for data forwarding.

This model allows for calculating the optimal paths and forwarding strategies within the network, focusing on maximizing data packet reception at base stations and minimizing delays. The cooperative strategy among UAVs can be integrated into this model, enhancing efficiency and reliability in dynamic conditions that are particularly relevant for environmental monitoring and disaster management applications.

4. Experimental results and discussions

The provided graph in Fig. 2. Illustrates the Packet Delivery Ratio (PDR) performance of four different data forwarding strategies in a UAV-assisted WSN over a series of steps: GreedyForwarding, NoForwarding, QLearningForwarding, and RandomForwarding, each represented by distinct colours and markers. The GreedyForwarding strategy, shown by the blue line, demonstrates a high and relatively stable PDR, quickly reaching and maintaining a ratio slightly above 0.95, indicating effective packet delivery despite higher delays. The NoForwarding strategy, represented by the orange line, consistently has a PDR of zero throughout all steps, as no data forwarding occurs, resulting in no packet deliveries. The QLearningForwarding strategy, depicted by the green line, starts with a low PDR but shows a steady and significant increase over time, eventually approaching 0.9. This upward trend highlights the effectiveness of the Q-Learning algorithm in improving delivery success rates as agents learn and optimize their forwarding strategies. The RandomForwarding strategy, shown by the red line, exhibits a moderate PDR, starting around 0.5 and gradually increasing to just below the GreedyForwarding performance, slightly above 0.9, indicating that while random decisions can lead to reasonably successful deliveries, they are not as consistently effective as learned strategies. The QLearningForwarding strategy's significant improvement over time underscores the algorithm's success in balancing exploration and exploitation, refining strategies for better outcomes. GreedyForwarding maintains a high and stable PDR, suggesting reliability in ensuring packet delivery. At the same time, QLearningn Forwarding, despite an initial learning curve, demonstrates substantial gains, eventually matching or surpassing the effectiveness of simpler approaches like GreedyForwarding. This highlights the robustness of the Q-Learning approach in optimizing data forwarding in UAV-assisted WSNs.

Fig. 3. Illustrates the delay performance of four different data forwarding strategies in a UAV-assisted WSN across various episodes: GreedyForwarding, NoForwarding, QLearningForwarding, and RandomForwarding, each represented by distinct colours and markers. The GreedyForwarding strategy, shown by the blue line, maintains a consistently high delay of around 105 units throughout the episodes, indicating that this strategy while ensuring high packet delivery, results in significant delays due to likely congestion and inefficient routing paths. The NoForwarding strategy, represented by the orange line, shows a constant delay of zero, as expected because no data is forwarded. Hence, no delay is incurred, although this is impractical for real-world applications requiring data transmission. The QLearning-Forwarding strategy, depicted by the green line, exhibits a lower delay than the others, stabilizing around 45 units, demonstrating the Q-Learning algorithm's effectiveness in optimizing data forwarding routes and minimizing delays through continuous learning and adaptation. The RandomForwarding strategy, shown by the red line, results in a moderate delay, fluctuating around 60 units, indicating that while random forwarding can sometimes find efficient paths, it generally leads to higher delays due to the lack of systematic optimization. The QLearningForwarding strategy stands out by achieving a significantly lower and more stable delay, underscoring the success of the Q-Learning approach in balancing exploration and exploitation to refine data forwarding policies. Comparatively, GreedyForwarding and RandomForwarding, although simpler, incur higher delays, making them less suitable for applications where minimizing delay is critical. This graph highlights the advantages of using the Q-Learning algorithm in UAV-assisted WSNs to optimize data forwarding, reduce delays, and improve overall network performance.

Fig. 4. Illustrates the delay performance of the QLearningForwarding strategy within a UAV-assisted Wireless Sensor Network (WSN) over a series of episodes. The vertical axis represents the delay in specific units, while the horizontal axis corresponds to the number of episodes,

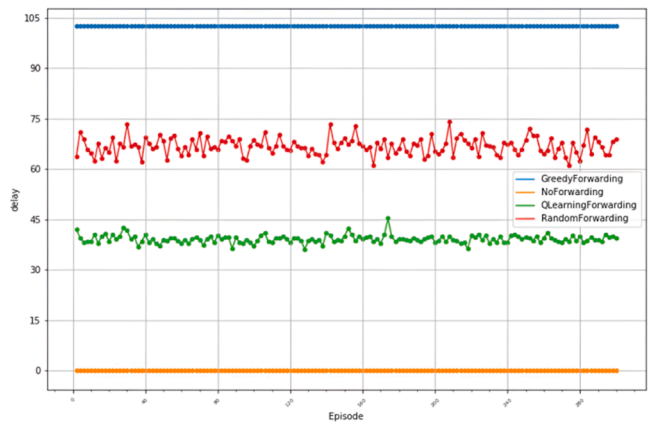

Fig. 3. Comparison of delay performance for different data forwarding strategies in a UAV-assisted WSN across episodes, highlighting the effectiveness of the Q-Learning approach.

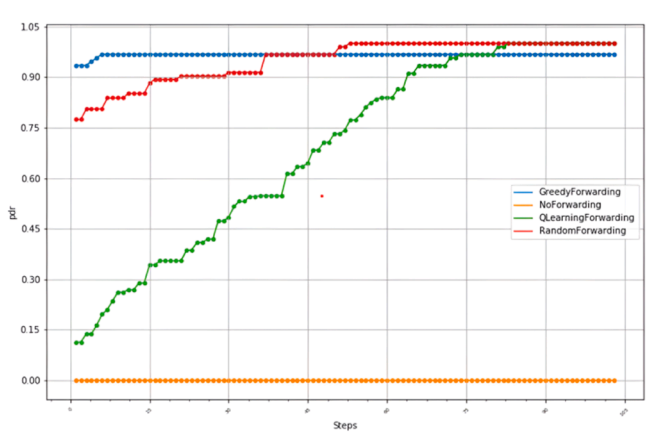

Fig. 2. Comparison of PDR for different data forwarding strategies in a UAV-assisted WSN over time, highlighting the Q-Learning approach.

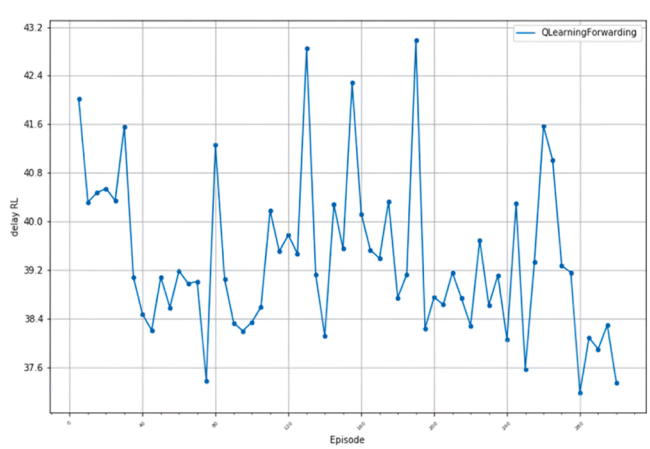

Fig. 4. Delay performance of the QLearningForwarding strategy over episodes.

reflecting the temporal progression of the training process.

The delay values are initially relatively high, ranging from approximately 41 to 42 units. This phenomenon is anticipated as the Q-Learning algorithm begins operating with a high exploration rate. During this phase, the algorithm prioritizes exploring various potential paths to accumulate sufficient information about the environment, inevitably resulting in suboptimal routing decisions and higher delays. As the training advances, the delay values exhibit notable variability, characterized by pronounced peaks and troughs. This variability is indicative of the inherent exploration-exploitation trade-off in the Q-Learning process. Specifically, the peaks in delay are associated with episodes where the algorithm explores less optimal or previously untested paths. At the same time, the troughs correspond to episodes where the algorithm capitalizes on known, effective paths.

Towards the later stages of training, a downward trend in delay values becomes evident, with the minimum delay approaching approximately 37.6 units. Concomitantly, there is a discernible reduction in variability, with delay values converging towards a narrower range of around 39 units. This convergence suggests that the Q-Learning algorithm has effectively learned and refined its routing strategy, transitioning from exploration to a more stable exploitation phase. The reduced delay values and variability highlight the algorithm's success in optimizing data forwarding decisions, ultimately enhancing network performance. The graph further underscores the learning efficiency of the QLearningForwarding strategy. The reduction in delay over successive episodes demonstrates the UAVs' progressive acquisition of experience and ability to make increasingly informed routing decisions. The convergence of the delay values towards the final episodes provides strong evidence of the algorithm's ability to identify and adopt optimal policies for data forwarding, thereby minimizing communication delays in the UAV-assisted WSN.

The decreased delay over the training episodes is primarily attributed to the Q-Learning algorithm's ability to refine its routing strategy through iterative learning. Initially, higher delays occur due to the algorithm exploring suboptimal paths to gather environmental data. However, as the training progresses, the algorithm leverages its accumulated knowledge to identify and exploit more efficient paths, reducing delays. This improvement is facilitated by the algorithm's gradual shift from exploration to exploitation, where it prioritizes proven optimal routing decisions. Consequently, the decreasing delay signifies the algorithm's convergence towards a stable and efficient policy, optimizing data forwarding in the UAV-assisted WSN.

Fig. 5. Compares the performance of various data forwarding strategies in a UAV-assisted Wireless Sensor Network (WSN) context. Specifically, it evaluates their impact on delay over a series of steps. The delay metric is critical in UAV-assisted WSNs, as it directly affects the timeliness and reliability of information dissemination, particularly in scenarios where data freshness is paramount, such as disaster response,

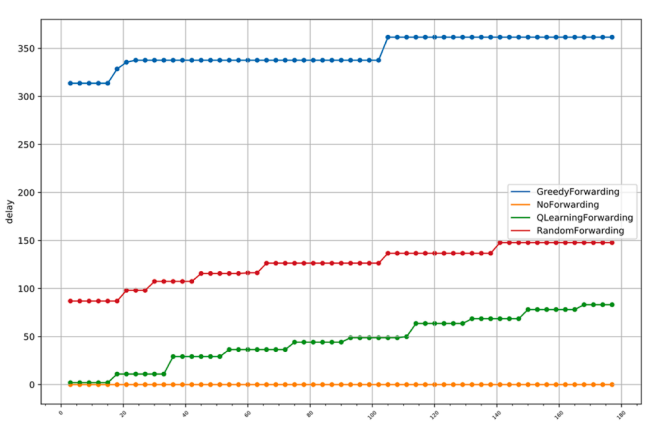

Fig. 5. Illustrate the compression of delay with a multi-based station.

environmental monitoring, or military operations. The blue curve, representing Greedy Forwarding, demonstrates the highest delay among all strategies, starting at approximately 300 and increasing steadily to around 350 over the steps. This consistent increase in delay underscores the inherent limitations of Greedy Forwarding. Although this method prioritizes immediate or short-term benefits by choosing the next hop that seems most direct, it often fails to account for the broader network conditions, such as potential congestion at bottleneck nodes. This lack of foresight results in cumulative latency, as traffic congestion exacerbates delays over time. Consequently, while Greedy Forwarding may initially appear efficient, its inability to anticipate and mitigate network congestion leads to significant performance degradation in dynamic environments like UAV-assisted WSNs.

The orange curve, indicative of the NoForwarding strategy, remains flat at the bottom of the graph, signalling negligible delay throughout the observed steps. While this suggests an absence of delay, the lack of data forwarding implies a failure of the network to perform its fundamental function—data dissemination. This strategy is impractical in real-world applications, as it effectively halts communication within the network. While the absence of delay might seem favourable at first glance, it must be understood that this comes at the cost of total functionality, rendering the network ineffective for practical purposes.

The green curve, corresponding to the Q-Learning Forwarding strategy, showcases a gradual but steady increase in delay, starting near zero and culminating around 70. This performance highlights the advantages of reinforcement learning-based approaches, where the system continuously learns from its environment to make optimal routing decisions. By dynamically adjusting its strategy based on accumulated experience, Q-Learning Forwarding effectively balances network load and minimizes congestion. The comparatively low delay associated with Q-Learning reflects its ability to adapt to the dynamic and often unpredictable conditions of UAV-assisted WSNs. This adaptability is particularly valuable in such networks, where the frequent movement of UAVs can lead to constant changes in network topology. The results underscore the potential of reinforcement learning to significantly enhance delay performance, making it a promising approach for optimizing UAV-assisted WSNs.

The red curve, representing Random Forwarding, depicts a moderate and progressively increasing delay, stabilizing around 150. The lack of a structured approach to route selection results in inefficiencies, as packets are forwarded without considering network conditions or traffic loads. While Random Forwarding performs better than Greedy Forwarding in terms of delay, its stochastic nature prevents it from achieving the optimized performance observed with Q-Learning. Compared to Q-Learning, the higher delay highlights the need for more intelligent and adaptive routing strategies in dynamic networks.

The figure clearly illustrates that static and simplistic forwarding strategies, such as Greedy Forwarding and Random Forwarding, fall short in dynamic network environments like UAV-assisted WSNs. Greedy Forwarding's myopic focus on immediate gains and Random Forwarding's lack of systematic routing result in suboptimal delay performance. On the other hand, Q-Learning Forwarding demonstrates the potential of adaptive and intelligent algorithms to mitigate delay, optimize network performance, and enhance the overall efficiency of UAV-assisted WSNs.

The delay observed in the various forwarding strategies can be attributed to differences in how each approach manages network congestion, route selection, and adaptability to changing topologies. Greedy Forwarding accumulates significant delay due to its focus on immediate, locally optimal decisions, often leading to congestion at bottleneck nodes. While avoiding strict patterns, Random Forwarding suffers from inefficiency and lack of optimization, resulting in moderate delays. In contrast, Q-learning forwarding exhibits lower delays due to its adaptive nature, dynamically learning from network conditions to optimize routes and distribute traffic effectively. The No Forwarding strategy shows zero delay, but this is due to the absence of actual data

dissemination, making it impractical in real-world applications. These variations highlight the critical role of intelligent and adaptive algorithms in minimizing delay and improving network performance in dynamic environments like UAV-assisted WSNs.

The Fig. 6, provides highlights the comparative analysis of Packet Delivery Ratio (PDR) for four distinct forwarding strategies: Greedy Forwarding, No Forwarding, Q-Learning Forwarding, and Random Forwarding in the context of UAV-assisted Wireless Sensor Networks (WSNs). This visualization encapsulates the dynamic evolution of PDR across a series of steps, reflecting the effectiveness of these strategies in reliably delivering data packets to their intended destinations.

The greedyforwarding approach, represented by the blue curve, demonstrates a rapid increase in PDR, stabilizing near 0.9. This trend underscores its efficacy in consistently delivering packets by leveraging geographic information. The strategy operates by selecting the next node closest to the destination, which minimizes the hop distance at every stage. Such a method aligns well with geographic routing principles, prioritising efficient route selection to reduce latency and improve reliability. The high PDR achieved suggests that the strategy is robust, even in dynamic environments typical of UAV-assisted WSNs, where mobility introduces variability in network topology. The red curve, indicative of the RandomForwarding strategy, starts with a lower initial PDR than GreedyForwarding but gradually improves, levelling off at approximately 0.75. This method, characterized by the random selection of forwarding nodes, inherently involves a degree of unpredictability. While this randomness may contribute to suboptimal routing paths and occasional inefficiencies, the strategy's performance suggests that it can still achieve moderate success in packet delivery. The eventual stabilization of the PDR reflects a balance between the inherent randomness and the network's capacity to tolerate and adjust to such deviations over time.

The green curve corresponds to the QLearningForwarding strategy, which begins with a low initial PDR but shows a steady upward trajectory, ultimately reaching around 0.6. This improvement illustrates the adaptive capabilities of reinforcement learning, where Q-learning algorithms iteratively refine their decision-making processes based on feedback from the network. The gradual rise in PDR highlights the algorithm's ability to learn optimal routes through exploration and exploitation. However, the plateauing at a lower PDR than the GreedyForwarding and RandomForwarding strategies may indicate challenges such as suboptimal convergence, insufficient exploration of better routes, or difficulty adapting to highly dynamic network conditions. Despite these challenges, the QLearningForwarding approach showcases potential in environments where static routing strategies fail to adapt to changing conditions.

The NoForwarding strategy, represented by the orange curve, predictably remains flat at a PDR of 0 throughout the steps. This serves as a baseline comparison, emphasizing the importance of active forwarding mechanisms for achieving any level of data delivery. The absence of packet forwarding inherently results in a complete lack of delivery, highlighting the critical role of effective forwarding strategies in UAV-assisted WSNs.

The increase in PDR for the QLearningForwarding strategy can be attributed to its adaptive learning mechanism, which iteratively improves routing decisions based on feedback from the network environment. Initially, the PDR is low due to the exploration phase, where the algorithm tries various paths to gather information. Over time, as the algorithm identifies and exploits more efficient routes, the PDR steadily increases. This improvement reflects the strategy's ability to learn optimal forwarding decisions by balancing exploring new routes and exploiting known, successful ones. However, the slower convergence compared to another strategy may result from challenges dealing with UAV-assisted networks' dynamic and uncertain conditions, emphasizing the need to fine-tune the exploration-exploitation trade-off.

Fig. 7 illustrates the decay of the epsilon value over a series of steps, a common approach in reinforcement learning where epsilon represents the exploration-exploitation trade-off in algorithms like epsilon-greedy. Initially, the epsilon value is high, indicating more exploration, but as the number of steps increases, epsilon gradually decreases, allowing the algorithm to exploit the learned policy with less exploration. In the context of UAV-assisted WSNs, this decay curve can represent the adaptive behaviour of the UAVs over time. In the early stages, UAVs explore potential actions, such as selecting flight paths or sensor nodes to optimize communication and energy efficiency. As the UAVs learn optimal routes and communication strategies, exploration decreases, and they focus on exploiting the most effective actions. This reinforcement learning-based approach allows UAVs to efficiently manage network resources and minimize energy consumption in WSNs by learning from dynamic environmental conditions.

5. Performance comparison with literature

Recent research has advanced several significant approaches for optimizing UAV-assisted data collection in IoT networks. Our comparative analysis evaluates our NSGA-II and Q-learning hybrid approach against three established methodologies. [28] proposed an energy-efficient trajectory optimization scheme that employs successive convex approximation (SCA) to minimize the weighted sum of UAV and sensor energy consumption through joint optimization of UAV trajectory, data collection time, and sensor wake-up scheduling. [29]introduced the SPS-IUTO scheme (matrix completion-based Sampling Points Selection joint Intelligent UAV Trajectory Optimization), combining spatial-temporal correlation-based matrix completion with ant colony

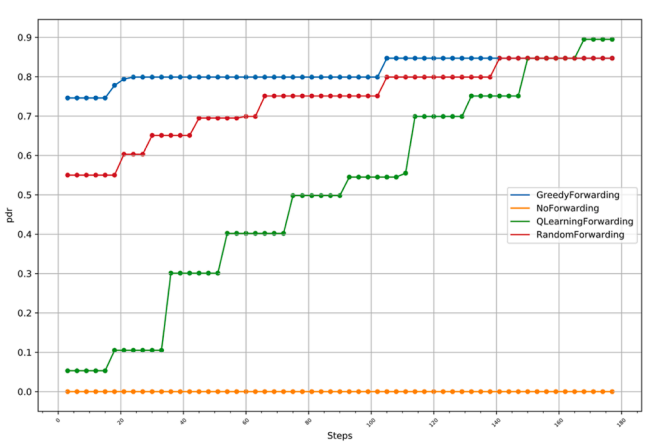

Fig. 6. Represent a comparative of PDR in UAV-assessed WSN in multi-UAV and base stations.

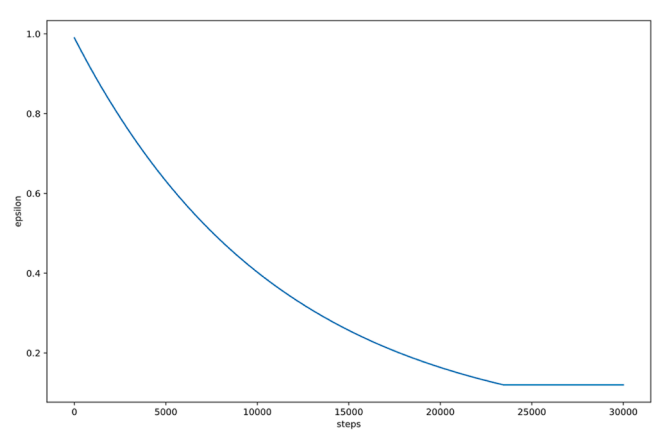

Fig. 7. Epsilon Decay Curve Demonstrating the Balance Between Exploration and Exploitation Over Time.

optimization for trajectory planning. Their method achieved a 17 % extension in network lifetime and a 50 % reduction in data redundancy compared to random sampling approaches. [30] developed a deep reinforcement learning methodology for UAV-assisted data collection, demonstrating improved packet delivery ratio (PDR) and delay performance compared to traditional greedy and random forwarding methods, particularly in scenarios with varying coverage radius. Our proposed hybrid approach demonstrates superior performance in terms of packet delivery ratio (PDR) and delay metrics compared to random forwarding and greedy forwarding methods. When the coverage radius R varies between 2 and 4 normalized grid lengths, our approach maintains a PDR above 0.95, random forwarding achieves approximately 0.9 PDR, and greedy forwarding shows high but unstable performance. Regarding the delay, our approach stabilizes around 45 units compared to 105 units for greedy forwarding and 60 units for random forwarding, demonstrating more efficient data collection through optimization of both UAV trajectory and transmission scheduling.

6. Conclusion

This research presents a comprehensive framework for optimizing UAV-assisted data collection and routing in IoT sensor networks, addressing critical challenges such as energy efficiency, coverage, and communication latency by decomposing the problem into two interdependent sub-problems—UAV deployment and data routing; our approach offers a practical solution that combines multi-objective optimization and reinforcement learning techniques.

By applying NSGA-II and NSGA-III algorithms, we successfully optimize UAV deployment patterns to ensure maximal coverage, reduced energy consumption, and minimal operational costs. Integrating Q-Learning for dynamic routing further enhances the system's performance by enabling adaptive decision-making, ensuring efficient data transmission even in evolving and unpredictable network environments. Problem decomposition has allowed us to focus on both UAV deployment and routing strategies independently while maintaining interdependence, significantly improving overall network performance.

Experimental evaluations have demonstrated that the proposed framework outperforms traditional strategies regarding energy consumption, data delivery delay, and PDR. The simulations showed that reinforcement learning-based routing significantly reduces latency compared to greedy and random forwarding methods. At the same time, the multi-objective optimization of UAV deployment offers superior energy savings and coverage capabilities. These findings underscore the potential of combining optimization algorithms and machine learning techniques to enhance the efficiency of UAV-assisted IoT networks.

The proposed methodology holds great promise for practical applications in various fields, including environmental monitoring, disaster response, and precision agriculture, where UAVs can gather data from vast and complex areas while maintaining low operational costs. Future research directions could focus on enhancing the solution's scalability for more extensive, complex networks and exploring integrating advanced machine learning models to predict network conditions and optimize routing decisions.

In conclusion, the framework presented in this research offers a novel, robust, and efficient solution for UAV-assisted data collection and routing in IoT sensor networks. The approach addresses key operational challenges by leveraging multi-objective optimization and reinforcement learning, providing a pathway to more sustainable and scalable IoT network deployments.

CRediT authorship contribution statement

Yasir I. Mohammed: Writing – original draft, Validation, Methodology, Investigation, Conceptualization. **Mohammad Kamrul Hasan:** Writing – original draft, Methodology, Investigation, Data curation, Conceptualization. **Rosilah Hassan:** Writing – original draft, Software, Methodology, Investigation, Conceptualization. **Muhammad Attique Khan:** Writing – review & editing, Software, Resources, Methodology, Data curation. **Huda Saleh Abbas:** Writing – review & editing, Validation, Software, Investigation. **Deepak Gupta:** Writing – review & editing, Supervision, Project administration, Investigation, Formal analysis. **Jamel Baili:** Writing – review & editing, Validation, Project administration, Methodology, Funding acquisition.

Declaration of Competing Interest

All authors declared no conflict of interest in this work. All authors contributed equally in this work. Thanks

Acknowledgments

This work has been supported by the Universiti Kebangsaan Malaysia, Under GUP 2023-010 and the research work has been conducted at the Network and Communication Technology Lab. The authors extend their appreciation to the Deanship of Research and Graduate Studies at King Khalid University for funding this work through Large Research Project under grant number RGP.2/379/45.

Appendix

See Table 1.

Table 1
List of abbreviation

Item	Descriptions
UAV	Unmanned Arial Vehicles
IoT	Internet of Things
LIDAR	Light Detection and Ranging
3D	Three Dimensions
NSGAII	Non-Dominated Sorting Genetic Algorithm- II
WSNs	Wireless Sensor Networks
TSP	Traveling Salesman Problem
MILP	Mixed-Integer Linear Programming
LSAP	Linear Sum Assignment Problem
DC	Difference of Convex
SCA	Successive Convex Approximation
FTP	Fine-Grained Trajectory Plan
FEB	Forwarding Engine Board

(*continued on next page*)

Table 1 (*continued*)

Item	Descriptions
PDR	Packet Delivery Ratio
MDVRPLS	Multi-Depot Vehicle Routing Problem with Limited Supply
MAC	Medium Access Control
FANET	Flying Ad Hoc Networks
MIMO	Multiple-Input Multiple-Output
SDMA	Space-Division Multiple Access
AoI	Age of Information

References

[1] K. Messaoudi, O.S. Oubbati, A. Rachedi, A. Lakas, T. Bendouma, N. Chaib, A survey of UAV-based data collection: Challenges, solutions and future perspectives, J. Netw. Comput. Appl. 216 (2023) 103670.

[2] R. Chengoden, et al., Metaverse for healthcare: a survey on potential applications, challenges and future directions, IEEE Access 11 (2023) 12765–12795.

[3] X. Xiao, X. Wang, and W. Lin, "Joint AoI-Aware UAVs Trajectory Planning and Data Collection in UAV-Based IoT Systems: A Deep Reinforcement Learning Approach," *IEEE Transactions on Consumer Electronics*, 2024.

[4] M.A. Hassan, S.I. Ullah, A. Salam, A.W. Ullah, M. Imad, F. Ullah, Energy efficient hierarchical based fish eye state routing protocol for flying ad-hoc networks, Indones. J. Electr. Eng. Comput. Sci. 21 (1) (2021) 465–471.

[5] J. Yoon, Y. Kim, S. Lee, M. Shin, UAV-based automated 3D modeling framework using deep learning for building energy modeling, Sustain. Cities Soc. 101 (2024) 105169.

[6] V. Saillaja, M. Menaka, V. Kumaravel, K. Machap, Development of an IoT-based Inventory Management System for Retail Stores. in *2023 International Conference on Sustainable Computing and Smart Systems (ICSCSS)*, IEEE, 2023, pp. 954–958.

[7] A.A. Habib, M.K. Hasan, R. Hassan, S. Islam, R. Thakkar, N. Vo, Distributed denial-of-service attack detection for smart grid wide area measurement system: a hybrid machine learning technique, Energy Rep. 9 (2023) 638–646.

[8] N. Gunantara, A review of multi-objective optimization: methods and its applications, Cogent Eng. 5 (1) (2018) 1502242.

[9] R. Hassan, Z. Daud, S. Usman, Internet of Things for Smart Solar Energy: An IoT Farm Development. in *2022 International Conference on Business Analytics for Technology and Security (ICBATS)*, IEEE, 2022, pp. 1–5.

[10] L. Li, et al., Unmanned aerial vehicles towards future Industrial Internet: Roles and opportunities, Digit. Commun. Netw. 10 (4) (2024) 873–883.

[11] R. Hassan, F. Qamar, M.K. Hasan, A.H.M. Aman, A.S. Ahmed, Internet of Things and its applications: a comprehensive survey, Symmetry 12 (10) (2020) 1674.

[12] A. Beishenalieva and S.-J. Yoo, "UAV Path Planning for Data Gathering in Wireless Sensor Networks: Spatial and Temporal Substate-Based Q-Learning," *IEEE Internet of Things Journal,* 2023.

[13] M. Bilal, Z. Hassan, M. Ayub, I. Ahmad, R. Odarchenko, Classification of Routing Protocols for FANTS. in *2019 IEEE International Scientific-Practical Conference Problems of Infocommunications, Science and Technology (PIC S&T)*, IEEE, 2019, pp. 350–358.

[14] M.A. Azzawi, R. Hassan, K.A.A. Bakar, A review on Internet of Things (IoT) in healthcare, Int. J. Appl. Eng. Res. 11 (20) (2016) 10216–10221.

[15] Y. Qiu, L. Ma, R. Priyadarshi, Deep learning challenges and prospects in wireless sensor network deployment, Arch. Comput. Methods Eng. (2024) 1–24.

[16] X. Xu, Z. Lin, X. Li, C. Shang, Q. Shen, Multi-objective robust optimisation model for MDVRPLS in refined oil distribution, Int. J. Prod. Res. 60 (22) (2022) 6772–6792.

[17] J. Chen, J. Wang, J. Wang, L. Bai, Joint fairness and efficiency optimization for CSMA/CA-based multi-user MIMO UAV ad hoc networks, IEEE J. Sel. Top. Signal Process. (2024).

[18] D. Huang, Z. Zhang, X. Fang, M. He, H. Lai, B. Mi, STIF: a spatial–temporal integrated framework for end-to-end micro-UAV trajectory tracking and prediction with 4-d mimo radar, IEEE Internet Things J. 10 (21) (2023) 18821–18836.

[19] M. Hanzla, et al., UAV detection using template matching and centroid tracking, IEEE Access (2024).

[20] S. Goudarzi, N. Kama, M.H. Anisi, S. Zeadally, S. Mumtaz, Data collection using unmanned aerial vehicles for Internet of Things platforms, Comput. Electr. Eng. 75 (2019) 1–15.

[21] P. Consul, I. Budhiraja, D. Garg, N. Kumar, R. Singh, A.S. Almogren, A hybrid task offloading and resource allocation approach for digital twin-empowered UAV-assisted MEC network using federated reinforcement learning for future wireless network, IEEE Trans. Consum. Electron. (2024).

[22] X. Huang, Y. Zhang, Y. Qi, C. Huang, M.S. Hossain, Energy-EFFICient UAV scheduling and probabilistic task offloading for digital twin-empowered consumer electronics industry, IEEE Trans. Consum. Electron. (2024).

[23] C.-L. Hu, S.-Z. Huang, Z. Zhang, L. Hui, Energy-balanced optimization on flying ferry placement for data gathering in wireless sensor networks, IEEE Access 9 (2021) 70906–70923.

[24] X. Xiong, C. Sun, W. Ni, X. Wang, Three-dimensional trajectory design for unmanned aerial vehicle-based secure and energy-efficient data collection, IEEE Trans. Veh. Technol. (2022).

[25] M.B. Ghorbel, D. Rodríguez-Duarte, H. Ghazzai, M.J. Hossain, H. Menouar, Joint position and travel path optimization for energy efficient wireless data gathering using unmanned aerial vehicles, IEEE Trans. Veh. Technol. 68 (3) (2019) 2165–2175.

[26] C. Luo, M.N. Satpute, D. Li, Y. Wang, W. Chen, W. Wu, Fine-grained trajectory optimization of multiple UAVs for efficient data gathering from WSNs, IEEE/ACM Trans. Netw. 29 (1) (2020) 162–175.

[27] A.T. Azar, et al., Drone deep reinforcement learning: a review, Electronics 10 (9) (2021) 999.

[28] C. Zhan, R. Huang, Energy minimization for data collection in wireless sensor networks with uav. in *2019 IEEE Global Communications Conference (GLOBECOM)*, IEEE, 2019, pp. 1–6.

[29] X. Liu, Y. Liu, N. Zhang, W. Wu, A. Liu, Optimizing trajectory of unmanned aerial vehicles for efficient data acquisition: a matrix completion approach, IEEE Internet Things J. 6 (2) (2019) 1829–1840.

[30] M. Yi, X. Wang, J. Liu, Y. Zhang, B. Bai, Deep reinforcement learning for fresh data collection in UAV-assisted IoT networks. in *IEEE INFOCOM 2020-IEEE Conference on Computer Communications Workshops (INFOCOM WKSHPS)*, IEEE, 2020, pp. 716–721.